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EARLY DETECTION OF FINANCIAL STRESS USING TRANSACTION PATTERNS

A Data Mining Approach using :
i) Frequent Pattern Mining
ii) Clustering
iii) Anomaly Detection
iv) Classification

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ABSTRACT

This project investigates how personal financial behaviour can be analysed through data mining techniques to detect early indicators of financial stress. Using five years of real financial transactions, various analytical methods - including preprocessing, exploratory data analysis, frequent pattern mining, clustering, anomaly detection, and classification - were applied to uncover hidden patterns. The study demonstrates that personal spending behaviour contains clear trends, periodic cycles, and identifiable behavioural signatures.

A hybrid anomaly detection approach, combined with machine learning classifiers, was able to detect unusual financial periods with high accuracy. The results highlight the potential of data-driven methods in supporting personalised financial decision-making and stress monitoring.

Beyond identifying patterns, the project also emphasizes the value of feature engineering in capturing subtle behavioural signals that raw transactions cannot reveal on their own.

By transforming monthly spending into meaningful indicators such as volatility, essential - spending ratio, and category diversity, the analysis provides a richer view of financial habits. These features not only improved the quality of clustering and anomaly detection but also enabled the classification models to perform with exceptional accuracy.

Overall, the project demonstrates how even a single user's financial data can be transformed into actionable insights when supported by thoughtful data preparation and the right analytical techniques.

MOTIVATION

Managing personal finances is one of the biggest challenges faced by students relocating abroad. From fluctuating living expenses and currency conversions to rent, utilities, food, transportation, and unexpected costs, every month can look drastically different. As an international student in the United States, understanding how my financial behaviour evolves over time became essential for budgeting and planning. This project was therefore motivated by the need to analysing multi-year transaction history and uncover deeper insights that go beyond surface-level tracking. The goal was to understand patterns, identify irregularities, and explore whether data mining techniques could help in detecting months that may lead to financial stress.

In addition to personal relevance, this project also serves as an academic exploration of how real-world financial data can be transformed into meaningful behavioural indicators using data mining techniques. Unlike traditional budgeting tools that simply record expenses, this study focuses on uncovering hidden structures, behavioural cycles, and abnormal spending events that may not be obvious at first glance. By combining domain understanding with analytical methods such as clustering, anomaly detection, and classification, the project aims to demonstrate how data-driven approaches can support better decision-making and help individuals anticipate financial pressure before it becomes overwhelming.

DATASET DESCRIPTION

The dataset consists of approximately 1500 manually tracked financial transactions recorded between Jan 2020 and Dec 2024. Each entry contains the transaction date, category, type (income or expense), description, and amount. The data represents real-life spending patterns across categories such as Food & Drink, Rent, Utilities, Shopping, Travel, Entertainment, Investment, Health & Fitness, and Salary. Since the dataset contains transactions spread across several years, it captures fixed monthly obligations, variable lifestyle-based spending, and irregular high-value transactions. This makes it suitable for behavioural mining and stress-related pattern detection.

While the dataset is personal in nature, its structure closely resembles real consumer financial data, making it a realistic foundation for data mining experimentation. The mix of recurring monthly expenses and occasional spikes provides a diverse set of patterns to analyse. Because the data spans multiple years, it also captures seasonal effects, long-term shifts in behaviour, and changes in spending habits over time. This richness allows the project to explore not just isolated transactions, but broader financial behaviours and trends that can be linked to periods of stability or stress.

	Date	Transaction Description	Category	Amount	Type
0	2020-01-02	Score each.	Food & Drink	1485.69	Expense
1	2020-01-02	Quality throughout.	Utilities	1475.58	Expense
2	2020-01-04	Instead ahead despite measure ago.	Rent	1185.08	Expense
3	2020-01-05	Information last everything thank serve.	Investment	2291.00	Income
4	2020-01-13	Future choice whatever from.	Food & Drink	1126.88	Expense

Dataset link : *https://www.kaggle.com/datasets/ramyapintchy/personal-finance-data/data*

METHODOLOGY

The project follows a structured data mining workflow divided into eight phases. Each phase focuses on transforming raw financial data into meaningful behavioural insights, supported by statistical and machine learning techniques.

This phased approach ensures that each analytical step builds logically on the previous one, creating a strong foundation for the techniques applied later in the project. By moving from basic understanding to deeper modelling, the workflow gradually enhances the quality and interpretability of insights. This structured methodology not only improves analytical clarity but also mirrors real-world data mining practices, where careful preparation, exploration, and transformation are essential before applying advanced algorithms.

Phase 1: Data Collection and Understanding

The first phase involved gaining familiarity with the dataset by reviewing its structure, column types, and descriptive statistics. Basic distribution analysis revealed that transaction amounts were right-skewed, indicating many small purchases and a few large transactions. Categorical exploration provided insights into the most frequent spending categories. This stage ensured a clear understanding of the dataset before deeper processing.

Beyond basic inspection, this phase also helped validate whether the dataset was suitable for downstream modelling. Understanding the natural distribution of amounts, the frequency of transactions, and the spread across categories revealed early clues about spending behaviour. This preliminary insight ensured that later steps such as clustering and anomaly detection were grounded in a realistic interpretation of the data, reducing the risk of misleading or unstable results.

	Date	Amount
count	1500	1500.000000
mean	2022-07-15 21:09:07.199999744	1307.520913
min	2020-01-02 00:00:00	14.370000
25%	2021-04-08 00:00:00	629.340000
50%	2022-07-27 00:00:00	1156.285000
75%	2023-10-05 06:00:00	1712.932500
max	2024-12-29 00:00:00	4996.000000
std	NaN	982.283361

Basic Statistical Summary of Transactions

	Transaction Description	Category	Type
count	1500	1500	1500
unique	1500	10	2
top	Pressure activity defense detail.	Rent	Expense
freq	1	165	1222

Category and Transaction Type Overview

Phase 2: Data Cleaning & Preprocessing

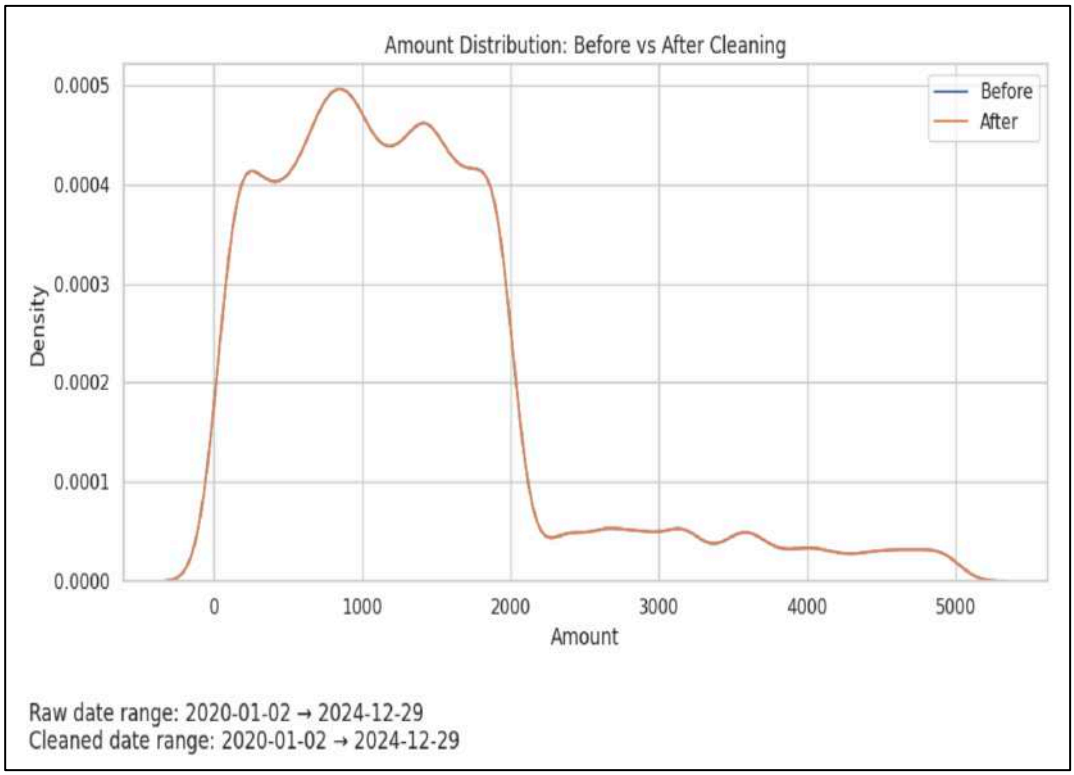
In the second phase, the dataset was cleaned and prepared for analysis. Duplicate records were removed, dates were converted into proper datetime format, and missing or inconsistent values were corrected. Several time-based features—such as Year, Month, Weekday, and Week Number - were engineered to support temporal analysis. Category labels were standardized to ensure consistency. This structured preprocessing provided a reliable foundation for exploratory analysis and modelling.

Preprocessing also played a crucial role in ensuring consistency across time. Converting dates into structured components such as month and week allowed the model to capture temporal

trends more effectively. Standardising categories and cleaning noisy entries ensured that algorithms treated similar transactions consistently. This phase not only improved the reliability of the dataset but also enabled the creation of advanced features needed for behavioural modelling.

	Date	Transaction Description	Category	Amount	Type	Year	Month	Day	Weekday	Week	YYYYMM
0	2020-01-02	Score each.	Food & Drink	1485.69	Expense	2020	1	2	Thursday	1	2020-01
1	2020-01-02	Quality throughout.	Utilities	1475.58	Expense	2020	1	2	Thursday	1	2020-01
2	2020-01-04	Instead ahead despite measure ago.	Rent	1185.08	Expense	2020	1	4	Saturday	1	2020-01
3	2020-01-05	Information last everything thank serve.	Investment	2291.00	Income	2020	1	5	Sunday	1	2020-01
4	2020-01-13	Future choice whatever from.	Food & Drink	1126.88	Expense	2020	1	13	Monday	3	2020-01

Cleaned and Standardized Dataset

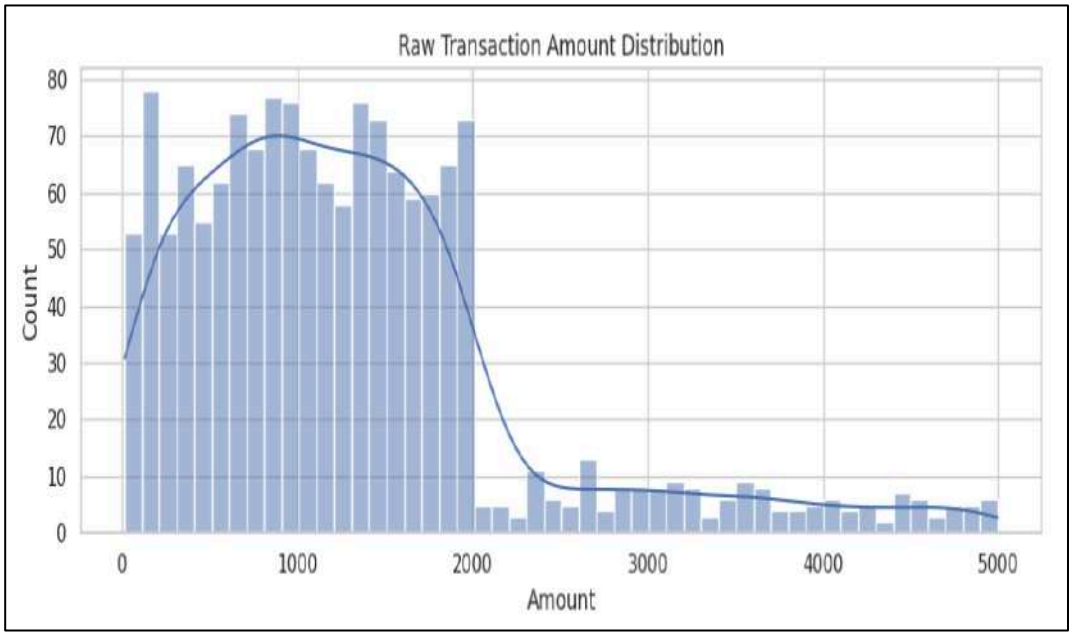


Effect of Data Cleaning on Amount Distribution

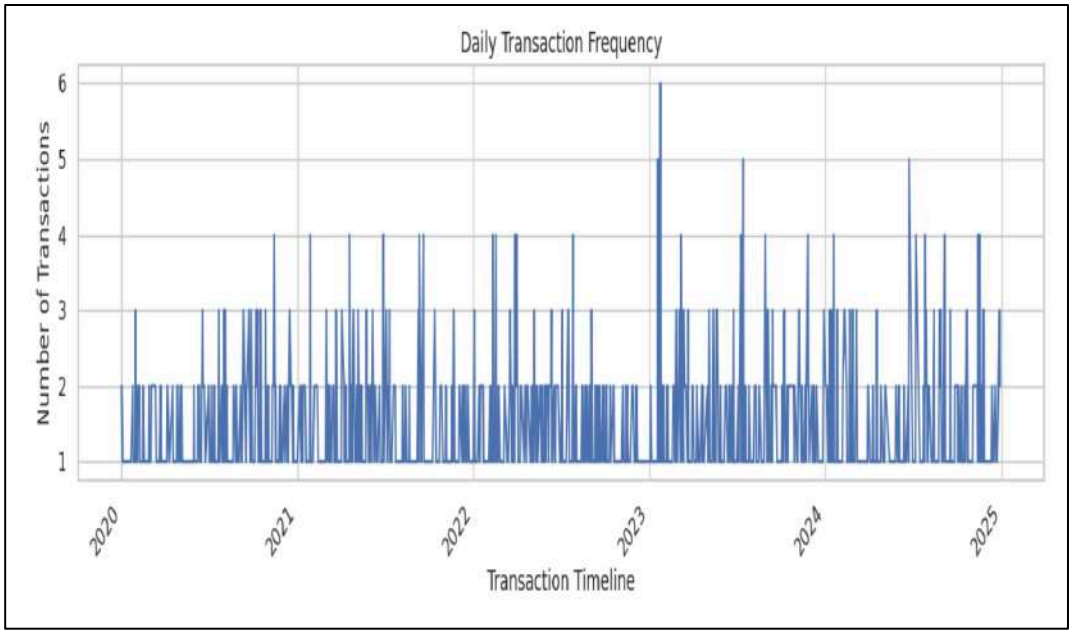
Phase 3: Exploratory Data Analysis (EDA)

Exploratory Data Analysis helped uncover important behavioural insights. Histogram and KDE plots showed the distribution of spending amounts, while line charts revealed monthly spending cycles and daily activity patterns. Category-based analysis highlighted which areas contributed most to expenses. Outlier detection methods such as the IQR technique identified unusual spending spikes. Seasonal heatmaps demonstrated how spending behaviour varied across months and weekdays. A correlation matrix was also used to evaluate relationships between numerical and temporal variables, helping guide later modelling steps.

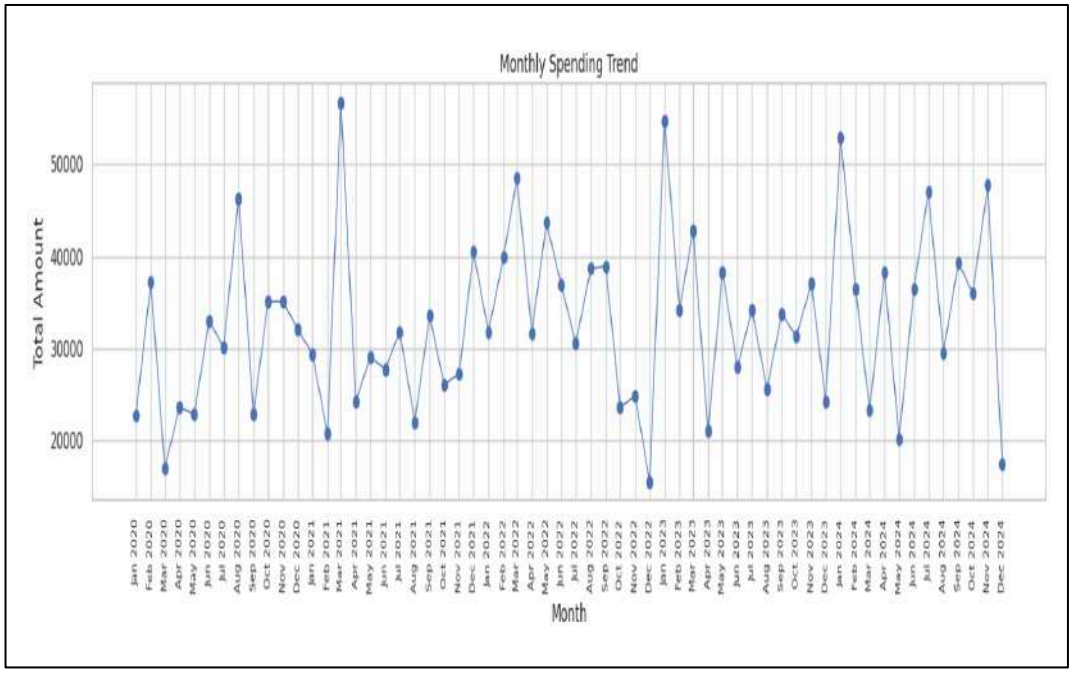
The patterns discovered in EDA helped shape the direction of later modelling steps. For example, the presence of clear monthly cycles suggested that time-based features would be valuable inputs to clustering and classification. Similarly, the detection of spending spikes supported the need for anomaly detection techniques. Overall, EDA served as both an exploratory and diagnostic stage, guiding feature engineering choices and highlighting behavioural characteristics that warranted deeper analysis.



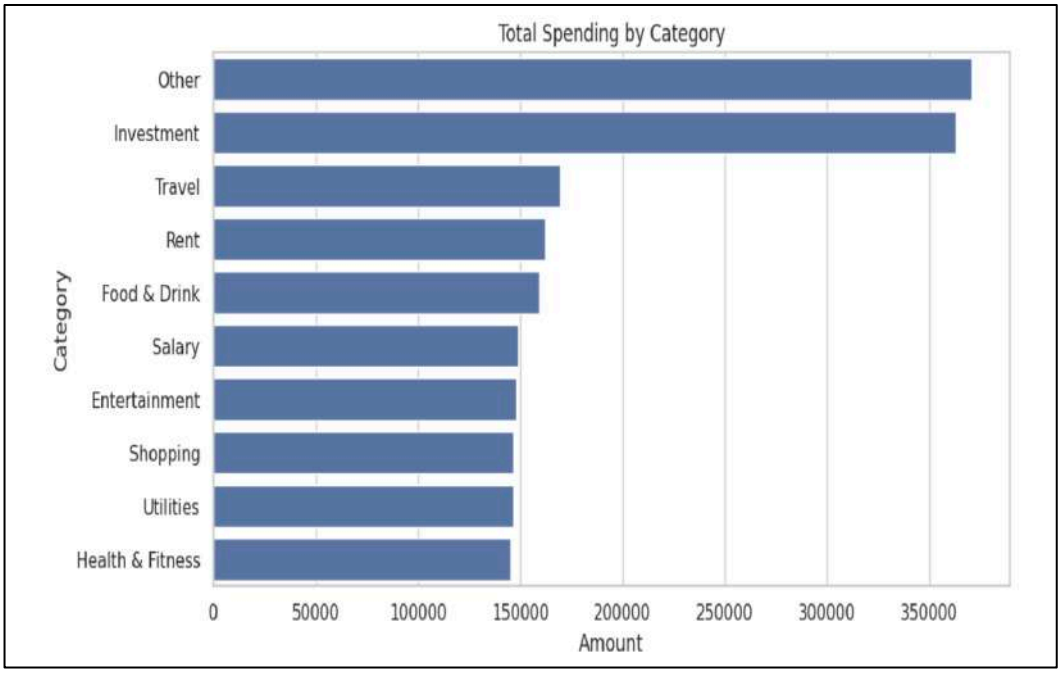
Distribution of Transaction Amounts
(Histogram + KDE)



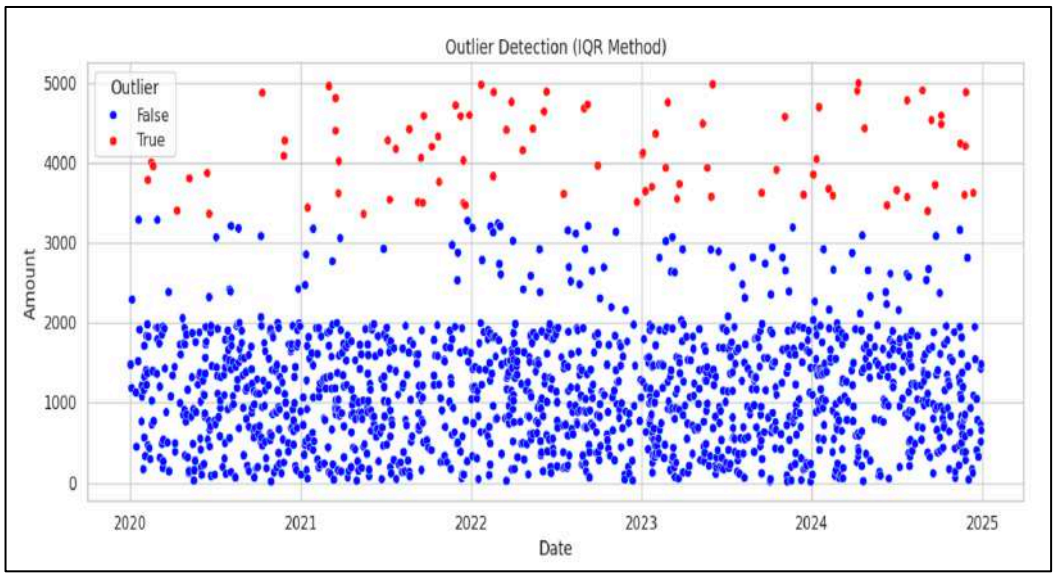
Daily Transaction Frequency Over Time



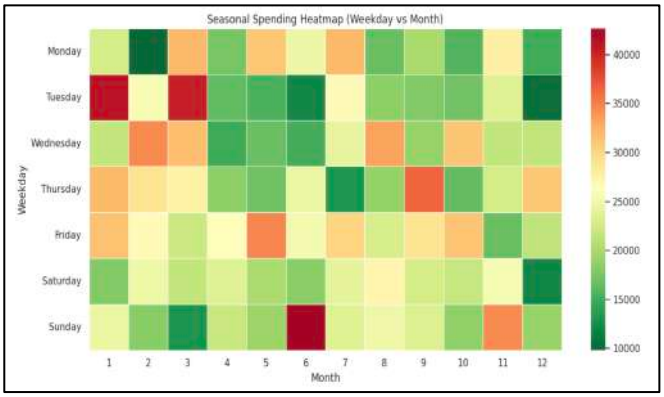
Monthly Spending Trend



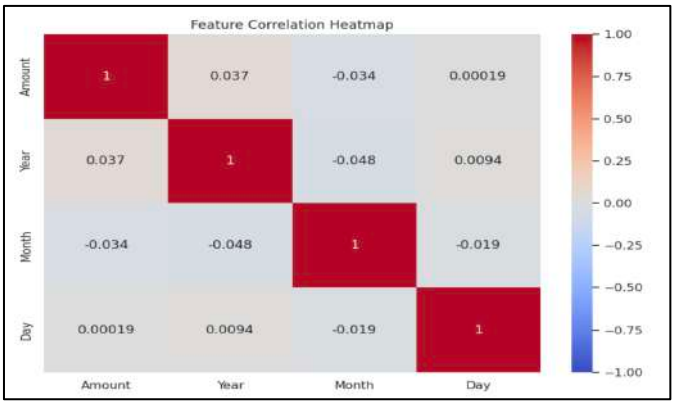
Total Spending by Category



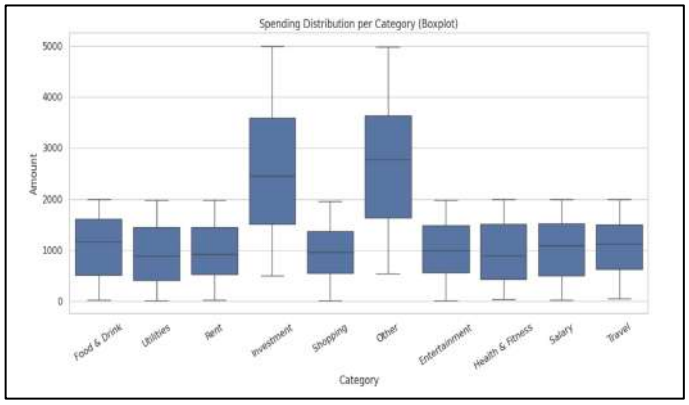
Outliers in Transactions (IQR Method)



Seasonal Spending Patterns (Weekday vs Month)



Feature Correlation Heatmap



Spending Distribution Across Categories

Phase 4: Feature Engineering

To transform raw transactions into meaningful behavioural metrics, new monthly features were engineered. These included total and average spending, volatility (standard deviation), maximum transaction value, skewness, kurtosis, transaction count, category diversity (entropy), and the ratio of essential to non-essential spending. These features captured both statistical characteristics and behavioural traits. They played a crucial role in clustering and classification, making financial patterns more interpretable and measurable.

These engineered features allowed the dataset to move beyond simple raw amounts and capture more nuanced behavioural signals. For instance, volatility and skewness helped differentiate stable months from highly irregular ones, while category entropy highlighted whether spending was concentrated or widely distributed. Together, these metrics created a multidimensional behavioural profile for each month, enabling algorithms to interpret spending behaviour more meaningfully.

A	B	C	D	E	F	G	H	I	J	K	L	M
UserID	total_spend	avg_monthly_spend	max_spend	min_spend	spend_volatility	avg_weekly_txn	growth_rate	pct_essential	category_diversity	amount_skew	amount_kurtosis	median_mean_ratio
User_1	1961281.37	1307.520913	4996	14.37	982.283361	28.301887	-0.230995	0.415333	2.301024	1.458602	2.461035	0.884334
Saved: user_features.csv												

User-Level Behavioural Features : Aggregated Financial Behaviour Across the Entire Timeline

Phase 5: Frequent Pattern Mining

Frequent Pattern Mining was carried out using Apriori and FP-Growth to discover common combinations of spending categories within months. Each month was treated as a “basket” of categories, and both algorithms generated frequent itemsets and association rules. Metrics such as support, confidence, lift, and conviction were used to evaluate rule strength. Interestingly, both algorithms produced identical patterns, although Apriori ran slightly faster due to the

dataset's small size. This phase helped identify recurring habits such as combinations of Utilities, Grocery, and Food expenses appearing together.

The insights from frequent pattern mining provided a deeper understanding of routine financial behaviour. Identifying which categories routinely occur together helped highlight predictable and consistent habits, such as months with combined utility, food, and health-related expenses. These recurring combinations also offered a behavioural baseline, making deviations more visible during anomaly detection. Thus, frequent pattern mining supported both interpretability and context-setting for later phases.

	YYYYMM	Category
0	2020-01	[Food & Drink, Utilities, Rent, Investment, Fo...
1	2020-02	[Health & Fitness, Travel, Rent, Investment, T...
2	2020-03	[Rent, Entertainment, Food & Drink, Rent, Trav...
3	2020-04	[Travel, Utilities, Shopping, Salary, Other, S...
4	2020-05	[Travel, Salary, Food & Drink, Health & Fitnes...

Monthly baskets table

	Entertainment	Food & Drink	Health & Fitness	Investment	Other	Rent	Salary	Shopping	Travel	Utilities
YYYYMM										
2020-01	1	1	1	1	1	1	1	1	0	1
2020-02	1	1	1	1	1	1	1	1	1	1
2020-03	1	1	1	1	1	1	0	1	1	1
2020-04	1	1	1	1	1	0	1	1	1	1
2020-05	0	1	1	1	0	1	1	1	1	1

One-hot encoded basket matrix

support	itemsets
0.950000	(Salary)
0.933333	(Entertainment)
0.933333	(Investment)
0.933333	(Health & Fitness)
0.933333	(Utilities)

FP-Growth top itemsets table

support	itemsets
0.950000	(Salary)
0.933333	(Health & Fitness)
0.933333	(Utilities)
0.933333	(Entertainment)
0.933333	(Investment)

Apriori top itemsets table

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	antecedents	consequents	antecedent sup	consequent sup	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
7288	(Utilities, Travel, Entertainment, Health & Fi...	(Other, Food & Drink)	0.633333	0.766667	0.5667	0.894737	1.16705	1	0.081111	2.216667	0.390374	0.68	0.548872	0.816934
7351	(Other, Food & Drink)	(Utilities, Travel, Entertainment, Health & Fi...	0.766667	0.633333	0.5667	0.73913	1.16705	1	0.081111	1.405556	0.613445	0.68	0.288538	0.816934
2497	(Entertainment, Utilities, Health & Fitness, S...	(Other, Food & Drink)	0.733333	0.766667	0.65	0.886364	1.15613	1	0.087778	2.053333	0.50641	0.764706	0.512987	0.867095
2516	(Other, Food & Drink)	(Entertainment, Utilities, Health & Fitness, S...	0.766667	0.733333	0.65	0.847826	1.15613	1	0.087778	1.752381	0.578755	0.764706	0.429348	0.867095
11410	(Utilities, Travel, Entertainment, Health & Fi...	(Other, Food & Drink)	0.583333	0.766667	0.5167	0.885714	1.15528	1	0.069444	2.041667	0.322581	0.62	0.510204	0.779814

FP-Growth association rules table

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	antecedents	consequents	antecedent sup	consequent sup	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
8150	(Utilities, Travel, Entertainment, Health & Fi...	(Other, Food & Drink)	0.633333	0.766667	0.566667	0.894737	1.167048	1	0.081111	2.216667	0.390374	0.68	0.548872	0.816934
8213	(Other, Food & Drink)	(Utilities, Travel, Entertainment, Health & Fi...	0.766667	0.633333	0.566667	0.73913	1.167048	1	0.081111	1.405556	0.613445	0.68	0.288538	0.816934
2927	(Entertainment, Utilities, Health & Fitness, S...	(Other, Food & Drink)	0.733333	0.766667	0.65	0.886364	1.156126	1	0.087778	2.053333	0.50641	0.764706	0.512987	0.867095
2946	(Other, Food & Drink)	(Entertainment, Utilities, Health & Fitness, S..	0.766667	0.733333	0.65	0.847826	1.156126	1	0.087778	1.752381	0.578755	0.764706	0.429348	0.867095
11730	(Utilities, Travel, Entertainment, Health & Fi...	(Other, Food & Drink)	0.583333	0.766667	0.516667	0.885714	1.15528	1	0.069444	2.041667	0.322581	0.62	0.510204	0.779814

Apriori association rules table

```
0.1 FP: 0.167 n_itemsets 175  | Apriori: 0.007 n_itemsets 175
0.15 FP: 0.09 n_itemsets 175  | Apriori: 0.007 n_itemsets 175
0.2 FP: 0.088 n_itemsets 175  | Apriori: 0.007 n_itemsets 175
```

Comparison output (runtime + number of itemsets)

Phase 6: Clustering Analysis

Clustering was used to group months into behavioural categories. K-Means clustering identified three meaningful spending clusters, while DBSCAN detected two dense groups and additional noise points. Principal Component Analysis (PCA) was used solely for visualisation to reduce dimensionality and display cluster boundaries. Evaluation metrics like Silhouette Score and Davies–Bouldin Index were used to assess the quality of clusters. Overall, K-Means produced more stable cluster boundaries, while DBSCAN was useful for identifying months with irregular behaviour.

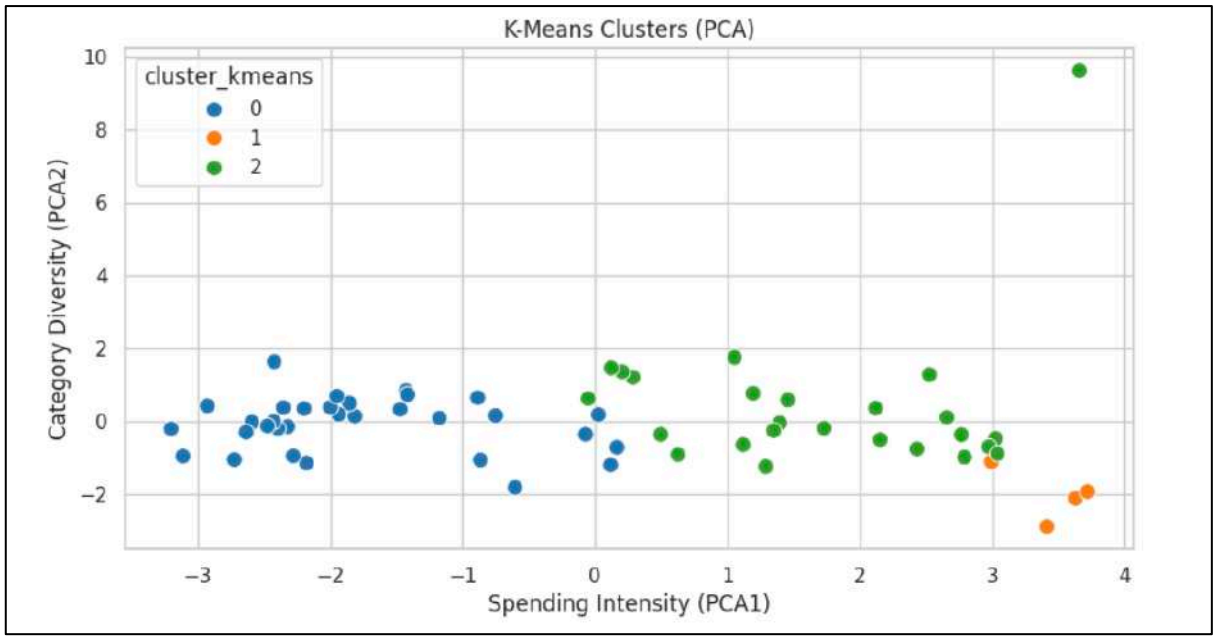
The combination of K-Means and DBSCAN ensured a more comprehensive understanding of behavioural clusters. While K-Means effectively captured broad spending patterns, DBSCAN revealed smaller, irregular clusters that represented unusual months. Using both techniques helped validate cluster robustness and allowed the study to distinguish between stable behaviours and more sporadic ones. This multi-model approach created a stronger foundation for anomaly detection, which benefits from contextualised cluster information.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
	YYYYMM	Amount_sum	Amount_mean	Amount_median	Amount_max	Amount_std	txn_count	category_entropy	pct_essential	spend_volatility	median_mean_ratio	amount_skew	amount_kurtosis
0	2020-01	12615.74	1146.885455	856.03	2420	653.683452	11	1.389681	0.181818	653.683452	0.746395	0.467425	-0.608001
1	2020-02	7732.34	1104.62	1475.58	1817.74	707.355973	7	1.549826	0.714286	707.355973	1.335826	-0.37774	-1.512332
2	2020-03	9799.36	1633.226667	1452.45	3071	988.065965	6	1.56071	0.333333	988.065965	0.889313	0.202725	-1.055184
3	2020-04	11148.52	1114.852	1193.49	3210	908.282489	10	1.973001	0.6	908.282489	1.070537	1.063267	0.919207
4	2020-05	16490.08	1268.467692	1337.12	2291	619.855351	13	2.138397	0.384615	619.855351	1.054122	-0.412871	-0.554906

Monthly Behavioural Feature Matrix

	PCA1	PCA2
Amount_sum	0.111163	0.519048
Amount_mean	0.333459	-0.284649
Amount_median	0.145159	-0.276083
Amount_max	0.466355	-0.012268
Amount_std	0.434822	-0.172314
txn_count	0.104499	0.525599
category_entropy	0.035288	0.323660
pct_essential	-0.103356	-0.025551
spend_volatility	0.434822	-0.172314
median_mean_ratio	-0.221443	-0.073663
amount_skew	0.335566	0.197815
amount_kurtosis	0.274336	0.295953

PCA Feature Contributions



K-Means Clusters (PCA Visualization)

Silhouette: 0.2703849063501059

DB Index: 1.3013239014281692

K-Means Cluster Quality Metrics

	YYYYMM
0	2020-01
1	2020-02
2	2020-03
4	2020-05
5	2020-06
8	2020-09
10	2020-11
11	2020-12
12	2021-01
15	2021-04
16	2021-05
18	2021-07
22	2021-11
23	2021-12
24	2022-01
25	2022-02
28	2022-05
31	2022-08
32	2022-09
33	2022-10
35	2022-12
36	2023-01
40	2023-05
41	2023-06
42	2023-07
43	2023-08
45	2023-10
46	2023-11
49	2024-02
50	2024-03
55	2024-08
56	2024-09

Cluster 0

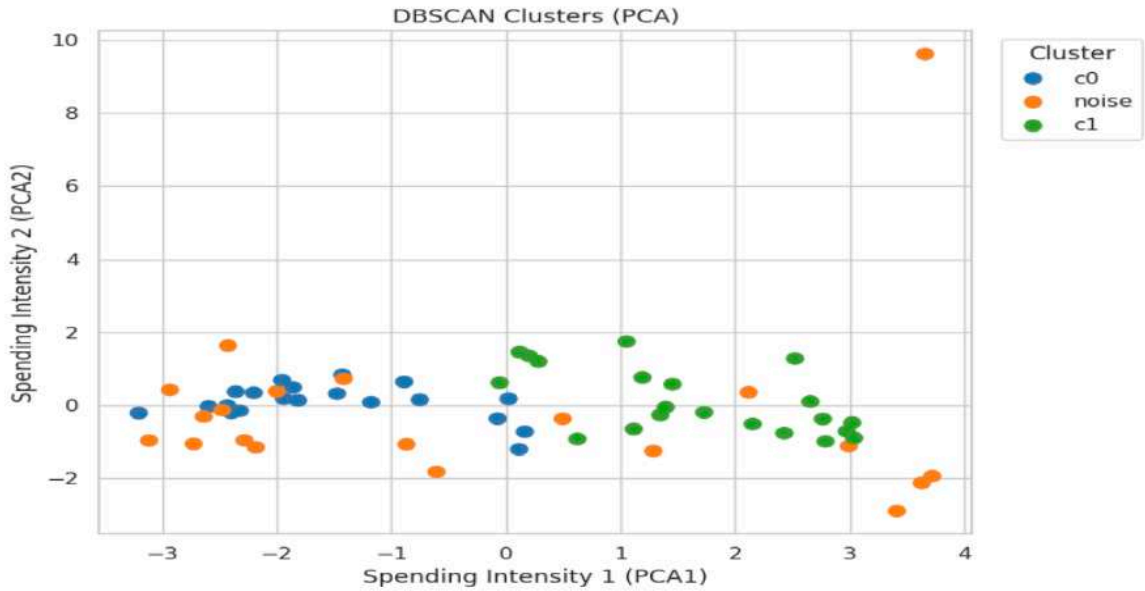
	YYYYMM
34	2022-11
37	2023-02
47	2023-12
51	2024-04

Cluster 1

	YYYYMM
3	2020-04
6	2020-07
7	2020-08
9	2020-10
13	2021-02
14	2021-03
17	2021-06
19	2021-08
20	2021-09
21	2021-10
26	2022-03
27	2022-04
29	2022-06
30	2022-07
38	2023-03
39	2023-04
44	2023-09
48	2024-01
52	2024-05
53	2024-06
54	2024-07
57	2024-10
58	2024-11
59	2024-12
60	NaT

Cluster 2

```
Cluster counts (including noise=-1):
cluster_dbscan
-1    20
 0    20
 1    21
Number of DBSCAN clusters (excluding noise): 2
Silhouette Score (non-noise subset): 0.412
Davies-Bouldin Index (non-noise subset): 0.9774
```



DBSCAN Cluster Evaluation Metrics and PCA visualisation

```
   method  n_clusters  noise_ratio  silhouette  davies_bouldin
0  kmeans           3        0.000      0.3393          0.8475
1  dbscan           4        0.131      0.2171          2.2208
Saved window_features_with_clusters.csv and compare_kmeans_dbscan_summary.csv

KMeans cluster sizes:
cluster_kmeans
0    32
1    28
2     1

DBSCAN label counts:
cluster_dbscan
 2    23
 0    21
-1     8
 1     5
 3     4
```

Comparative Study : KMeans vs DBSCAN

Phase 7: Hybrid Anomaly Detection

A novel hybrid anomaly detection approach was implemented by applying Isolation Forest and One-Class SVM inside each cluster rather than across the entire dataset. This cluster-aware method allowed anomalies to be judged relative to similar behavioural months. Isolation Forest effectively detected spending spikes and rare events, while One-Class SVM captured subtle deviations. A combined hybrid score was then calculated to rank anomalies. The results successfully highlighted months where spending patterns were significantly irregular and potentially stress-inducing.

Applying anomaly detection within clusters, rather than across the entire dataset, created a more fair and context-aware method for identifying irregular months. A month that seems unusual globally may be completely normal within its own spending pattern cluster. By combining Isolation Forest and One-Class SVM, the hybrid approach captured both large anomalies and subtle deviations. This ensured that the anomaly scores reflected genuine behavioural irregularities rather than noise or seasonal shifts.

	Date	Amount	Category	cluster	isof_score	isof_pct	isof_topk	isof_abs
1285	2024-04-10	4996.0	Investment	1	1.000000	True	True	True
1016	2023-06-02	4981.0	Investment	0	1.000000	True	True	True
588	2022-01-22	4975.0	Other	0	0.998351	True	True	True
1389	2024-08-25	4906.0	Other	0	0.985195	True	True	True
610	2022-02-18	4883.0	Investment	0	0.967700	True	True	True
334	2021-03-02	4960.0	Other	1	0.956543	True	True	True
1281	2024-04-07	4900.0	Other	1	0.888077	True	True	True
795	2022-09-07	4727.0	Investment	0	0.879285	True	True	True
349	2021-03-16	4809.0	Other	1	0.868167	True	True	True
547	2021-11-28	4718.0	Other	0	0.866773	True	True	True
214	2020-10-10	4875.0	Investment	1	0.857344	True	True	True
713	2022-06-11	4888.0	Other	1	0.856085	True	True	True
784	2022-08-31	4683.0	Investment	0	0.851227	True	True	True
1212	2024-01-16	4696.0	Other	1	0.849973	True	True	True
1479	2024-11-26	4882.0	Investment	1	0.844989	True	True	True

Top Anomalies Detected by Isolation Forest (Sorted by ISOF Score)

	Date	Amount	Category	cluster	ocsvm_score	ocsvm_pct	ocsvm_topk	ocsvm_abs
1016	2023-06-02	4981.00	Investment	0	1.000000	True	True	True
1285	2024-04-10	4996.00	Investment	1	1.000000	True	True	True
588	2022-01-22	4975.00	Other	0	0.971339	True	True	True
334	2021-03-02	4960.00	Other	1	0.842744	True	True	True
1389	2024-08-25	4906.00	Other	0	0.680078	True	True	True
1150	2023-11-09	15.45	Entertainment	0	0.653332	True	True	True
1121	2023-10-03	19.29	Shopping	0	0.637041	True	True	True
1281	2024-04-07	4900.00	Other	1	0.619019	True	True	True
1195	2023-12-29	14.37	Entertainment	1	0.616646	True	True	True
231	2020-10-29	15.32	Utilities	1	0.612916	True	True	True
393	2021-04-30	25.38	Rent	0	0.611669	True	True	True
865	2022-12-13	28.48	Salary	0	0.598973	True	True	False
85	2020-05-16	28.54	Rent	0	0.598729	True	True	False
1413	2024-09-21	3725.00	Other	0	0.598612	True	True	False
610	2022-02-18	4883.00	Investment	0	0.598464	True	True	False

Top Anomalies Detected by One-Class SVM (Sorted by OCSVM Score)

	Date	Amount	YYYYMM	Cluster	ISOF_Score	ISOF_Flag	OCSVM_Score	OCSVM_Flag
795	2022-09-07	4727.0	2022-09	0	0.879285	True	0.234466	False
349	2021-03-16	4809.0	2021-03	1	0.868167	True	0.367995	False
547	2021-11-28	4718.0	2021-11	0	0.866773	True	0.222674	False
214	2020-10-10	4875.0	2020-10	1	0.857344	True	0.539695	False
713	2022-06-11	4888.0	2022-06	1	0.856085	True	0.579939	False
784	2022-08-31	4683.0	2022-08	0	0.851227	True	0.185181	False
1212	2024-01-16	4696.0	2024-01	1	0.849973	True	0.190312	False
1479	2024-11-26	4882.0	2024-11	1	0.844989	True	0.561096	False
1357	2024-07-23	4780.0	2024-07	1	0.838253	True	0.309075	False
641	2022-03-28	4762.0	2022-03	1	0.836677	True	0.277313	False

Transactions Flagged by Isolation Forest ONLY

	Date	Amount	YYYYMM	Cluster	ISOF_Score	ISOF_Flag	OCSVM_Score	OCSVM_Flag
1150	2023-11-09	15.45	2023-11	0	0.713666	False	0.653332	True
1121	2023-10-03	19.29	2023-10	0	0.701407	False	0.637041	True
231	2020-10-29	15.32	2020-10	1	0.779537	False	0.612916	True
393	2021-04-30	25.38	2021-04	0	0.635064	False	0.611669	True
865	2022-12-13	28.48	2022-12	0	0.611593	False	0.598973	True
85	2020-05-16	28.54	2020-05	0	0.610206	False	0.598729	True
1413	2024-09-21	3725.00	2024-09	0	0.605456	False	0.598612	True
1090	2023-08-27	2815.00	2023-08	0	0.508566	False	0.598180	True
1144	2023-10-30	2815.00	2023-10	0	0.508566	False	0.598180	True
896	2023-01-23	3700.00	2023-01	0	0.587359	False	0.598152	True

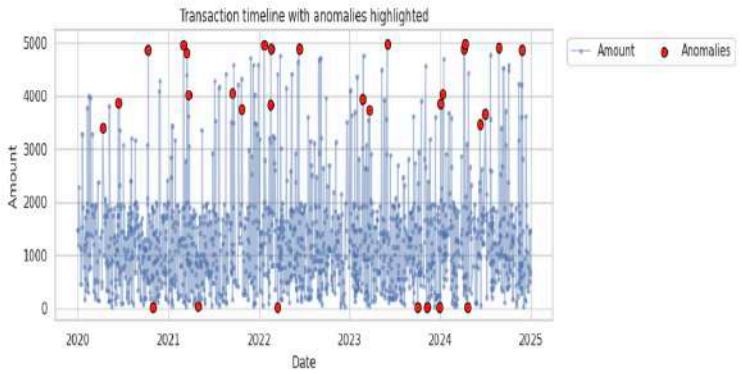
Transactions Flagged by One-Class SVM ONLY

```
===== GLOBAL ANOMALY DETECTOR COMPARISON =====
Total transactions: 1500
Isolation Forest flagged: 30  (0.0200)
One-Class SVM flagged:    30  (0.0200)
Both flagged:             8
Jaccard Similarity:       0.1538


===== PER-CLUSTER COMPARISON =====
```

	Cluster	n_txn	ISOF_Flagged	OCSVM_Flagged	Both_Flagged	ISOF_Ratio	OCSVM_Ratio	Jaccard
0	0	802.0	11.0	25.0	4.0	0.013716	0.031172	0.125
1	1	698.0	19.0	5.0	4.0	0.027221	0.007163	0.200

Hybrid Anomaly Detection: Algorithm Comparison
Isolation Forest vs One-Class SVM

```
Loaded hybrid_anomaly_results.csv
Warning: WF exists but lacks 'YYYYMM' or 'cluster' columns; skipping cluster merge.
Anomaly selection method: hybrid_flag | anomalies found: 30
```



Timeline chart

	Date	Amount	Category	YYYYMM	hybrid_score	isof_score	ocsvm_score	hybrid_flag
1286	2024-04-10	4996.00	Investment	2024-04	1.000000	1.000000	1.000000	True
1016	2023-06-02	4981.00	Investment	2023-06	1.000000	1.000000	1.000000	True
588	2022-01-22	4975.00	Other	2022-01	0.984845	0.998351	0.971339	True
336	2021-03-02	4960.00	Other	2021-03	0.899643	0.956543	0.842744	True
1389	2024-08-25	4906.00	Other	2024-08	0.832637	0.985195	0.680078	True
610	2022-02-18	4883.00	Investment	2022-02	0.783082	0.967700	0.598464	True
1281	2024-04-07	4900.00	Other	2024-04	0.753548	0.888077	0.619019	True
713	2022-06-11	4888.00	Other	2022-06	0.718012	0.856085	0.579939	True
1479	2024-11-26	4882.00	Investment	2024-11	0.703043	0.844989	0.561096	True
1195	2023-12-29	14.37	Entertainment	2023-12	0.698860	0.781075	0.616646	True

Combined Top 10 Hybrid Anomalies (Sorted by Hybrid Score)

Phase 8: Classification Modelling

Using anomaly-based labels, classification models were trained to predict high-risk months. Logistic Regression served as the baseline and performed well, but tree-based models such as Random Forest and XGBoost achieved perfect classification metrics. These models captured complex, non-linear behavioural patterns with ease, thanks to the strong engineered features. ROC and Precision-Recall curves confirmed that both Random Forest and XGBoost achieved complete separation between high-risk and low-risk months, making them highly reliable for stress prediction.

The comparison of models also highlighted the importance of selecting algorithms that align with the structure of the engineered features. While Logistic Regression provided valuable baseline performance, tree-based models such as Random Forest and XGBoost were far more capable of learning the non-linear relationships present in the behavioural data. This demonstrated that model choice has a significant impact on predictive accuracy and that carefully engineered features can greatly enhance classifier performance.

```
=== Logistic Regression ===
Train shape: (45, 14) Test shape: (16, 14)
Accuracy: 0.875
Precision: 0.8333
Recall: 0.8333
F1: 0.8333
ROC-AUC: 0.9667

Confusion Matrix:
[[9 1]
 [1 5]]

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90        10
           1       0.83      0.83      0.83         6

    accuracy                           0.88        16
   macro avg       0.87      0.87      0.87        16
weighted avg       0.88      0.88      0.88        16


Saved model: model_logistic.joblib summary: logistic_eval_summary.csv

Top features by |coefficient|:
```

	feature	abs_coef
0	0	1.131246
1	Amount_max	0.731570
2	pct_essential	0.731570
3	cluster_kmeans	0.697654
4	Amount_std	0.688783
5	Amount_sum	0.550051
6	Amount_median	0.518707
7	amount_kurtosis	0.299874
8	Amount_mean	0.287823
9	txn_count	0.186955

Logistic Regression - Model Performance Summary and Top Features by Absolute Coefficient

	feature	coef
0	0	1.131245
1	Amount_max	0.731570
2	pct_essential	0.731570
3	Amount_std	0.688783
4	Amount_sum	0.550051
5	Amount_median	0.518707
6	amount_kurtosis	0.299874
7	Amount_mean	0.287823

	feature	coef
13	cluster_kmeans	-0.697654
12	txn_count	-0.186955
11	spend_volatility	-0.089006
10	median_mean_ratio	0.103643
9	category_entropy	0.104926
8	amount_skew	0.126243
7	Amount_mean	0.287823
6	amount_kurtosis	0.299874

Logistic Regression - Positive Coefficients Driving Class 1 (High-Risk Months) and Negative Coefficients Driving Class 0 (Low-Risk Months)

```
=== Random Forest ===
Accuracy: 1.0 Precision: 1.0 Recall: 1.0 F1: 1.0 ROC-AUC: 1.0
Confusion matrix:
 [[10  0]
 [ 0  6]]

Classification report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         6

    accuracy                           1.00        16
   macro avg       1.00      1.00      1.00        16
weighted avg       1.00      1.00      1.00        16


Top features by importance (Random Forest):
```

	feature	importance
0	pct_essential	0.193986
1	0	0.166026
2	Amount_max	0.155731
3	Amount_median	0.117805
4	Amount_sum	0.093805
5	txn_count	0.055116
6	median_mean_ratio	0.053052
7	Amount_std	0.036489
8	amount_kurtosis	0.034977
9	amount_skew	0.024624

Random Forest - Model Performance Summary and Feature Importance Ranking

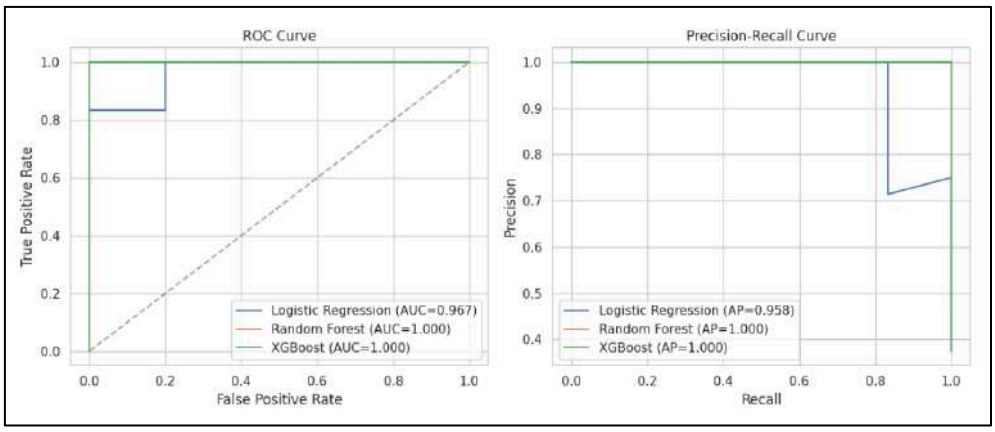
```
=== XGBoost ===
Train shape: (45, 14) Test shape: (16, 14)
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1: 1.0
ROC-AUC: 1.0

Confusion Matrix:
 [[10  0]
 [ 0  6]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         6

    accuracy                           1.00        16
   macro avg       1.00      1.00      1.00        16
weighted avg       1.00      1.00      1.00        16

Saved model: model_xgboost.joblib summary: xgboost_eval_summary.csv

Top features by gain (XGBoost):
```

	feature	gain
0	Amount_max	6.569448
1	0	2.059268
2	Amount_sum	0.166648
3	Amount_median	0.088317
4	cluster_kmeans	0.032856
5	median_mean_ratio	0.028116
6	txn_count	0.001930
7	spend_volatility	0.001247
8	amount_skew	0.000906

XGBoost - Model Performance Summary and Top Features by Gain

	Accuracy	Precision	Recall	F1_Score	ROC_AUC
Model					
Logistic Regression	0.875	0.8333	0.8333	0.8333	0.9667
Random Forest	1.000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.000	1.0000	1.0000	1.0000	1.0000

Classification Models:
Performance Metrics Comparison



ROC & Precision–Recall Curves: Model Comparison

CONCLUSION

This project successfully applied an end-to-end data mining pipeline to personal financial data. From understanding raw spending behaviour to engineering meaningful features, identifying patterns, forming behavioural clusters, detecting anomalies, and finally predicting stress levels, each phase contributed to a comprehensive behavioural analysis. The hybrid anomaly detection method provided nuanced insights, while classification models demonstrated strong predictive capabilities. Overall, the study highlights how personal finance data, when analysed correctly, can reveal hidden behavioural signatures and support informed decision-making.

Beyond the technical outcomes, this project also highlights the broader value of integrating data-driven thinking into personal finance. By transforming raw spending logs into structured behavioural insights, the study shows how individuals can better understand their habits, anticipate stress periods, and make more informed choices. The combination of statistical analysis, machine learning, and anomaly detection demonstrates that even simple personal datasets can reveal meaningful patterns when analysed thoughtfully. Ultimately, this project reinforces the idea that financial awareness is not only about tracking expenses - it is about understanding behaviour, adapting proactively, and using data as a tool for personal stability and growth.

FUTURE SCOPE

Beyond the technical outcomes, this project also highlights the broader value of integrating data-driven thinking into personal finance. When raw spending logs are transformed into structured behavioural insights, they reveal patterns that often remain invisible in day-to-day life. Individuals can understand how their habits shift across seasons, how certain categories dominate their expenses, and how irregular purchases influence their monthly budgets. These insights make it possible to anticipate periods of financial strain before they occur, rather than reacting afterward. The project demonstrates that even a modest personal dataset, when analysed with the right tools, can produce meaningful conclusions that support better decision-making.

Furthermore, the combination of machine learning, anomaly detection, clustering, and rule mining illustrates the power of a holistic analytical approach. These methods work together to uncover spending signatures, detect unusual financial behaviour, and differentiate between normal fluctuations and genuine risk indicators. This goes beyond tracking expenses—it provides a behavioural mirror that reflects financial discipline, volatility patterns, and lifestyle shifts over time. The project ultimately reinforces the idea that financial awareness is both a quantitative and psychological journey, where data becomes a tool for reflection, planning, and long-term personal stability.